



FUNDAMENTAL IN AI and ML PROJECT

TOPIC : STUDY FOCUS DETECTOR

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1. Introduction

This project is a simple **Study Focus Detector** created using **Python and OpenCV**. The main purpose of this system is to detect whether a student is **focused** or **distracted** while studying by monitoring body movement through a webcam.

The idea is very simple:

- If the student moves too much, the system assumes they are **distracted**.
- If the student stays mostly still, the system assumes they are **focused**.

This is a beginner-friendly project that introduces concepts like webcam capture, frame comparison, and motion detection.

2. Objectives

The main objectives of this project are:

- To access live webcam feed using OpenCV.
- To compare current and previous video frames.
- To detect motion based on pixel differences.
- To display a real-time message showing:

FOCUS and DISTRACTED

- To build a clean program structure by dividing the code into modules.

3. Working Principle

This system works by continuously reading frames from the webcam. Each frame is converted into **grayscale** because black-and-white images are easier and faster to process.

Then the system compares the current grayscale frame with the previous grayscale frame.

How Motion is Detected

- The difference between two frames is calculated.
- If there is movement, the difference will show bright/white pixels.

- A threshold is applied to make the difference image clearer (only 0 or 255).
- The system counts the white pixels.
- If the white pixels are too many, it means there is a lot of motion.

Final Decision

- If motion value is above the set limit → **DISTRACTED**
- Otherwise → **FOCUSED**

4. Modules Used

The program is divided into **5 separate modules** to keep the code clean and easy to manage.

Module 1: Camera Module

This module handles:

- Opening the webcam
- Capturing frames
- Converting frames to grayscale

Module 2: Movement Detection Module

This module handles:

- Comparing previous and current frames
- Finding the absolute difference
- Applying threshold
- Calculating motion level

Module 3: Focus Decision Module

This module decides the status:

- Focused
- Distracted
- Starting (first frame)

Module 4: Display Module

This module handles:

- Showing the output window
- Displaying text on the video feed
- Checking if the ESC key is pressed to exit

Module 5: Main Program

This is the main controller file.

It connects all modules and runs the complete program in a loop.

5. Technologies Used

The project uses:

- Python Programming Language
- OpenCV Library
- Webcam Camera Device
- Computer Vision Concepts

6. Output

The output is displayed as a webcam window titled:

"Study Focus Detector"

The screen shows a label:

- FOCUSED (Green)
- DISTRACTED (Red)
- STARTING (Yellow)

This output updates in real time.

7. Advantages

This project has many benefits:

- Easy to understand and implement
- Works in real-time
- Helps track movement during study
- Good beginner project for OpenCV
- Modular coding style improves readability

8. Limitations

Even though the project works well, it has some limitations:

- It only detects movement, not actual attention.
- Small light changes can sometimes be detected as motion.
- If a student is distracted but sitting still, it will still show "Focused".
- Threshold value may need adjustment for different environments.

9. Future Improvements

In the future, this project can be improved by adding:

- Face detection to ensure the student is present
- Eye detection to check if the student is looking at the screen/books
- AI-based focus detection using machine learning
- Logging focus time and distracted time into a file
- Adding sound alert when distracted

10. Conclusion

The Study Focus Detector is a useful and simple computer vision project that detects distraction based on motion. It demonstrates the basic working of frame comparison and motion detection using OpenCV. By dividing the program into 5 modules, the project becomes more organized and easier to maintain.

This project is a great starting point for students who want to learn computer vision and real-time video processing.